Faculty Research Grant & Publication Tracker

Lab 1 – Requirements Engineering & UML Use-Case Modelling

Problem Statement #09

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1. Problem Context

Right now, the Research Deanery tracks sponsored grants, faculty publication records, co-author sign-offs, and fund usage largely by hand — through spreadsheets, email chains, and a fair amount of manual cross-checking. The Faculty Research Grant & Publication Tracker is meant to bring all of that into a single portal: logging grant expenses, keeping publication metrics like indexing and citations current, routing co-author approvals, and giving the Dean a running view of how funds are being spent.

Target Actors

- Faculty Researcher – given in the problem statement
- Research Dean – given in the problem statement
- Research Administration Staff – added on top of the two given actors, since the lab requires at least three

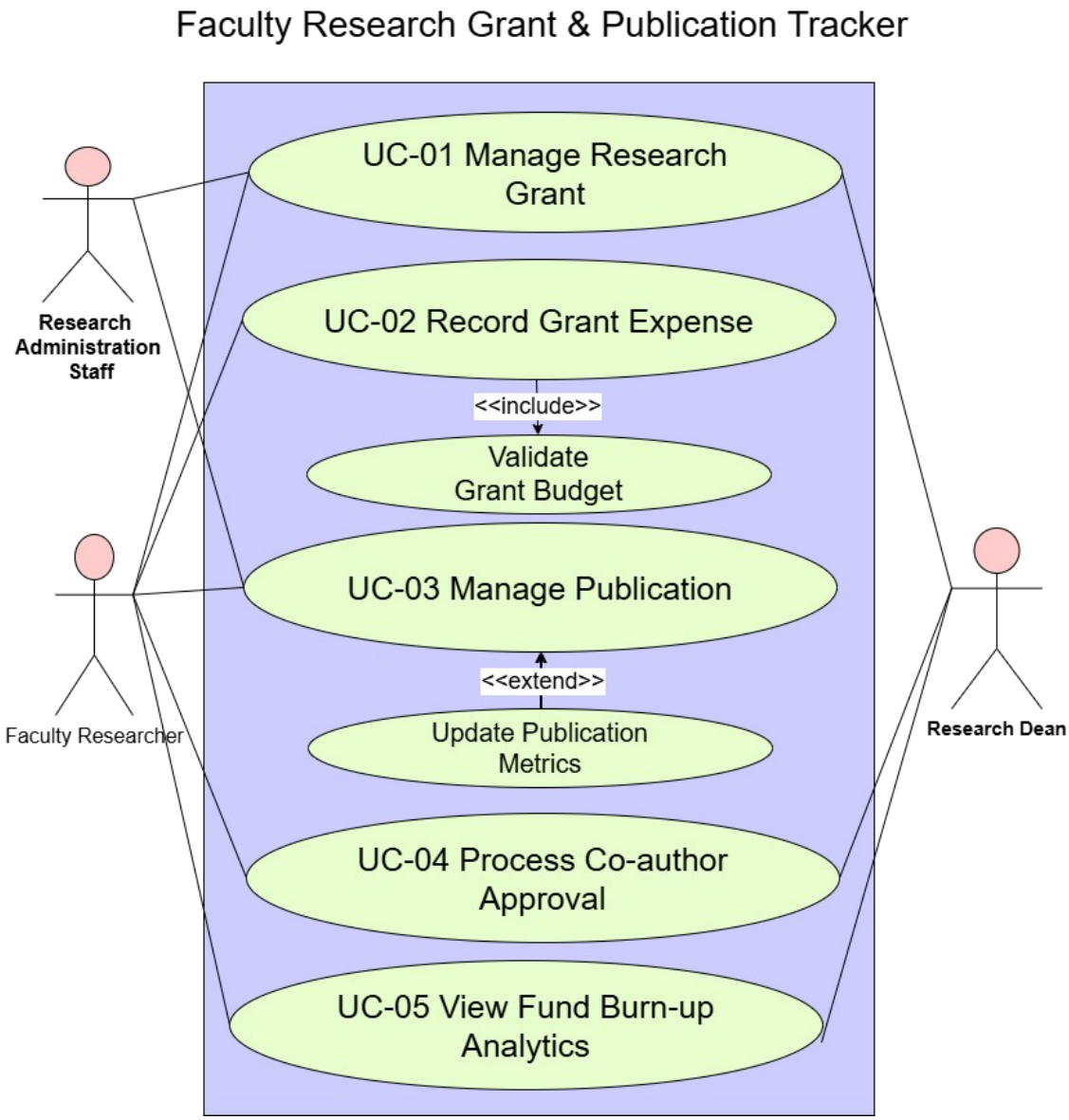
2. Requirements Table

Five functional requirements and two non-functional requirements were drawn from the problem statement below, each with an ID, type, description, priority, measurable acceptance criteria, and the reasoning behind it.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow faculty researchers to enter grant details, record expenses, and calculate the remaining grant amount.	High	Pass: The remaining amount is updated correctly after an expense is added, and an expense greater than the available amount is not accepted. Fail: The system allows spending beyond the available amount.	Grants get spent down through a series of small purchases, and it's easy to lose track of what's left once a few expenses pile up — this keeps the running balance accurate and stops faculty from overspending by accident.
FR-002	Functional	The system shall allow faculty researchers to add and update publication details such as title, journal/conference, indexing status, and citations.	High	Pass: A publication can be added with the required details and the updated information is displayed correctly. Fail: The publication cannot be saved or the details are displayed incorrectly.	Publication records are currently scattered across spreadsheets and emails; bringing them into one place makes it far easier for faculty to maintain an up-to-date list.
FR-003	Functional	The system shall allow the system to store and update the indexing and citation details of faculty publications.	High	Pass: Updated indexing or citation details are shown for the correct publication. Fail: Old or incorrect values are displayed.	Indexing and citation counts change after a paper is published, so the record needs to reflect that instead of staying frozen at the submission date.
FR-004	Functional	The system shall allow faculty researchers to submit co-author approval requests and allow the Research Dean to approve or reject them.	High	Pass: The request status changes to Approved or Rejected after the Dean takes action. Fail: The request status is not updated.	Co-authorship is currently settled over email threads with no clear record; a proper approval step gives both sides a traceable decision.
FR-005	Functional	The system shall show grant fund usage including total allocation, expenses, remaining amount, and spending trend.	High	Pass: The displayed allocation, expenses, and remaining amount match the stored grant records. Fail: The displayed values do not match the records.	The Dean needs a fast read on how a grant is being spent without pulling raw transaction logs every time.
NFR-001	Nonfunctional – Security	The system shall maintain an immutable audit record of financial approvals and journal status changes.	High	Pass: Every financial approval or status change is recorded with the user and time, and the record cannot be changed normally. Fail: A change occurs without being recorded.	Grant sponsors and internal audits both expect a record that can't be quietly edited after the fact.
NFR-002	Nonfunctional – Performance	The system shall display the fund burn-up analytics within 3 seconds for a normal request.	Medium	Pass: The analytics are displayed within 3 seconds in at least 95 out of 100 test requests. Fail: More than 5 requests take longer than 3 seconds.	A slow-loading analytics view is the kind of thing that quietly discourages people from checking it — keeping it under 3 seconds keeps it usable.

3. UML Use-Case Diagram

System: Faculty Research Grant & Publication Tracker



«include»: UC-02 Record Grant Expense → Validate Grant Budget — every time an expense is recorded, the budget check runs as part of it, so this is a mandatory include rather than a separate optional step.

«extend»: Update Publication Metrics → UC-03 Manage Publication — metrics like indexing and citations only get updated occasionally, when new data comes in, so this is modeled as an extend rather than something that happens every time a publication is managed.

4. Use-Case Flow Specification

Use Case: UC-02 – Record Grant Expense

Primary Actor: Faculty Researcher

Preconditions

1. Faculty Researcher is logged into the system.
2. An active research grant exists.
3. The researcher has permission to record expenses.

Postconditions

1. The expense is recorded successfully.
2. The remaining grant balance is updated.
3. The expense action is recorded in the audit ledger.

Main Success Scenario

1. Faculty Researcher picks an active research grant.
2. System shows the current grant balance.
3. Researcher selects “Record Expense.”
4. System opens the expense entry form.
5. Researcher fills in the expense details.
6. System validates the entered information.
7. System checks it against the available grant balance.
8. System confirms there are sufficient funds.
9. System records the expense.
10. System updates the remaining grant balance.
11. System logs the action in the audit ledger.
12. System displays the updated balance along with a confirmation.
13. Use case ends successfully.

Alternate Flow

7a. Expense Exceeds Available Grant Balance

1. System notices the expense is larger than the available grant balance.
2. System rejects the expense.
3. System shows an “Insufficient Grant Funds” message.
4. Researcher can either adjust the expense amount or cancel the operation.
5. If the amount is corrected, the flow picks back up from Step 7 of the main scenario.